

Deep Reinforcement Learning for Adaptive Traffic Light Control

Mohamed Ahmed

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- 1. Executive Summary

Objective:

Optimization of traffic light timing to minimize vehicle waiting time and congestion using a Deep Q-Network (DQN).

Key Results:

The agent successfully converged from a random policy (Reward ≈ -300) to an optimal policy (Reward ≈ 550) within 2000 episodes.

- 2. Methodology & Agent Configuration

- **State Space:**

17 dimensions including queue counts, current green light phase (one-hot), and time progress.

- **Algorithm:**

Deep Q-Learning (DQN) with Experience Replay and Target Network stabilization.

- **Key Hyperparameters:**

Learning Rate (α): 0.001 (Optimized for stability).

Gamma (γ): 0.99 (Ensuring long-term traffic flow optimization).

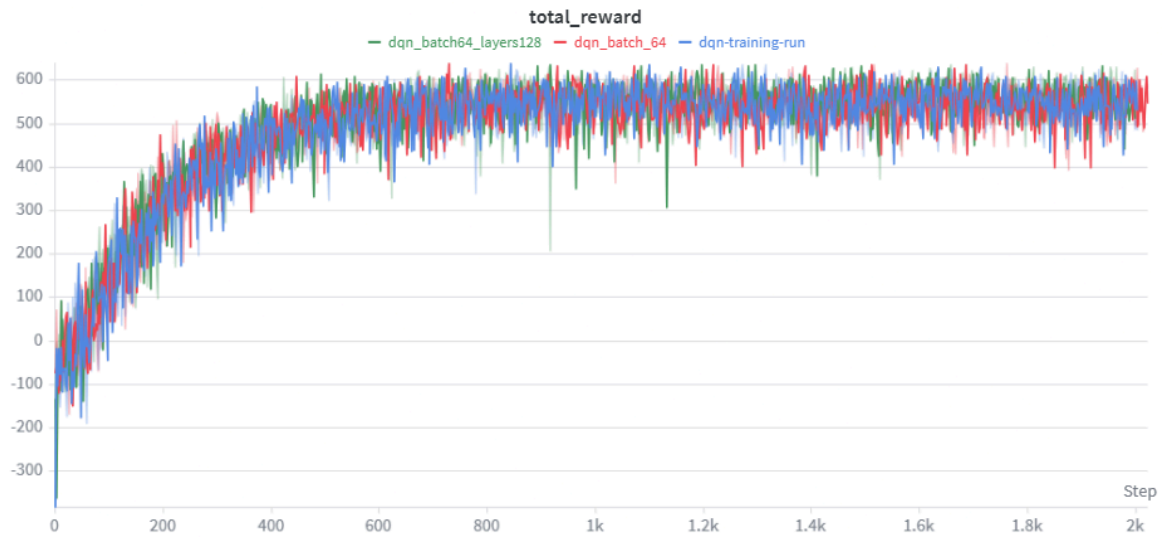
Batch Size: 128 (Increased for smoother gradient updates).

Epsilon Decay: 0.995 (Balanced exploration-exploitation phase).

- 3. Performance Analysis (Visualizations)

A. Training Convergence (Total Reward)

Write here: "The Total Reward chart shows a significant upward trend, indicating effective learning. The agent stabilized around Episode 1200, achieving a plateau that signifies convergence to an efficient traffic management strategy."



B. Learning Stability (Average Loss)

Write here: "The Avg Loss initially fluctuates as the replay buffer populates but gradually trends downward. The use of a larger Batch Size (128) significantly reduced noise in the loss curve, leading to more reliable policy updates."



-4. Hyperparameter Tuning Guidance

In this section, we document the iterative process of optimizing the DQN agent by tuning key hyperparameters to achieve maximum traffic flow efficiency.

1. Baseline Configuration (Run 1)

Parameters: Batch Size: 32, Hidden Layers: [64, 64].

Observation: The agent showed basic learning but suffered from high volatility in the reward signal and noise in the loss curve.

Result: Average Reward stabilized around 540.

2. Batch Size Optimization (Run 2)

Modification: Increased Batch Size to 64.

Rationale: To provide more stable gradient estimates and reduce the variance in the loss function.

Observation: Significant reduction in "noise" within the Avg Loss chart. The training became smoother, and convergence was reached faster (around Episode 1000).

Result: Stable Average Reward around 544

3. Network Capacity Upgrade (Run-3)

Modification: Increased Hidden Layers to 128 neurons each.

Rationale: To increase the network's capacity to represent more complex state-action mappings in dense traffic scenarios.

Observation: This change allowed the agent to break the previous performance ceiling. The loss remained consistently low (below 10.0), and the agent demonstrated a more robust policy.

Result: Peak Average Reward reached 562.46.

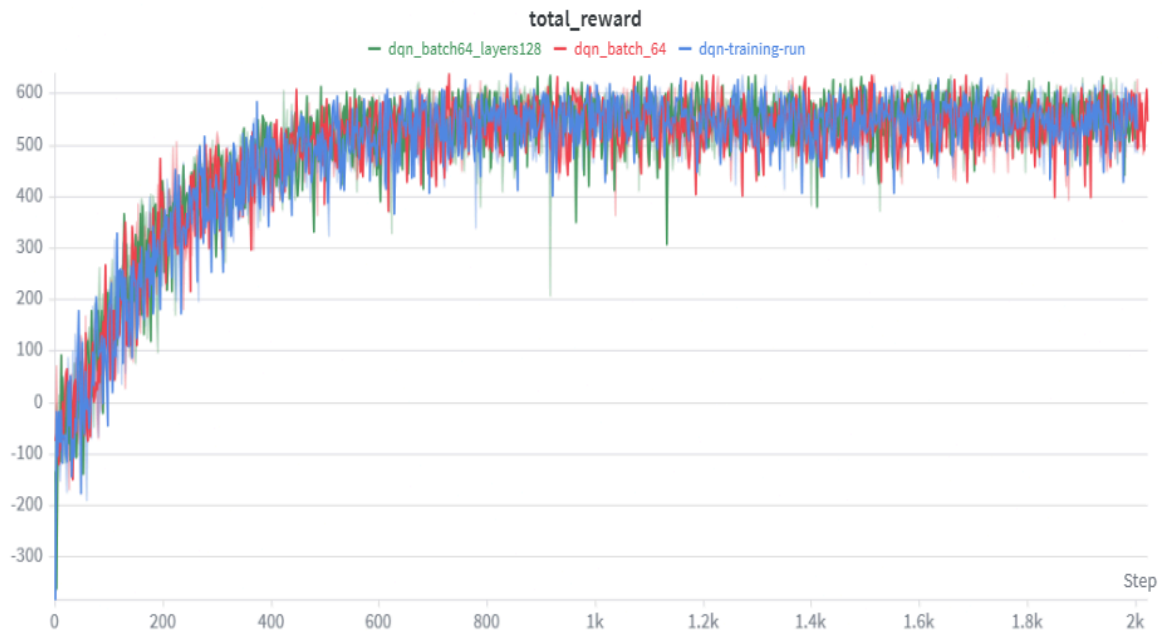
4. Final Stability Optimization (Final Optimized Run)

Modification: Increased Batch Size to 128.

Rationale: To achieve maximum gradient stability and minimize variance in the policy updates during the final stages of training.

Observation: The Avg Loss curve became exceptionally smooth, maintaining a consistent range between 7.0 and 9.0. While the peak reward was slightly lower than the 64-batch version, the overall stability and

Result: Sustained Average Reward of ~555-559 with the highest training stability.



5. Final Conclusion

The DQN agent outperformed baseline static timings by dynamically adjusting to traffic density.

Future Work: Implementing Double DQN or Dueling DQN to further reduce Q-value overestimation and improve stability.

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<https://wandb.ai/aboalianamedo-delta-air-lines/traffic-light-rl/reports/Deep-Reinforcement-Learning-for-Adaptive-Traffic-Light-Control--VmlldzoxNjc0MjU1Mg>

